

EPFL — College of Management of Technology | Financial Engineering

Financial applications of blockchains and distributed ledgers

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Module 1 — On-Chain Data Extraction

1. Volume Cross-Check

Each swap record in `swap_events.parquet` carries two signed token amounts: `amount0_usdc` (positive when USDC enters the pool, negative when it exits) and `amount1_weth` (same convention for WETH). By construction, a valid swap flips the sign between the two legs. We checked this for all 1,058,831 rows: no record has both amounts positive, both negative, or both zero. Every swap is a genuine exchange.

As a second check, we compared the USD notional computed from the USDC leg (`|amount0_usdc|`) against the notional implied by the WETH leg (`|amount1_weth| × price_usdc_per_weth`). The two agree to within floating-point rounding, with a mean relative deviation below 0.01%. This confirms that the `sqrPriceX96` decoding and decimal adjustments for both tokens are applied correctly throughout the dataset.

Over the six-month window, `buy_weth` trades account for 50.3% of all swaps and `sell_weth` for 49.7%, which is consistent with a balanced two-sided market.

2. Liquidity Map Spot-Check

The tick-level liquidity map in `liquidity_snapshots.parquet` is reconstructed by replaying every Mint and Burn event from pool deployment (block 12,376,729) up to each daily snapshot block. To check the replay logic, we validated the reconstructed state against the live chain at the first snapshot block of the study window: **block 23,479,243** (2025-10-01, $\approx 00:00$ UTC).

Methodology

Ten initialised ticks were sampled from the reconstructed map, spanning the full occupied range from tick $-887,270$ to tick $108,360$. For each tick, we issued a direct `eth_call` to `pool.ticks(tick_index)` on the Ethereum archive node at exactly block 23,479,243, retrieving the on-chain `liquidityGross` and `liquidityNet`. These were compared to the values produced by the event replay.

Results

Tick	Chain liqGross	Local liqGross	Chain liqNet	Local liqNet	Match
$-887,270$	2.308×10^{16}	2.308×10^{16}	2.308×10^{16}	2.308×10^{16}	✓
$-92,110$	3.983×10^{11}	3.983×10^{11}	3.983×10^{11}	3.983×10^{11}	✓
$-76,280$	2.965×10^{19}	2.965×10^{19}	2.965×10^{19}	2.965×10^{19}	✓
$-75,980$	2.965×10^{19}	2.965×10^{19}	-2.965×10^{19}	-2.965×10^{19}	✓
0	1.357×10^{13}	1.357×10^{13}	1.357×10^{13}	1.357×10^{13}	✓
100	100	100	100	100	✓
110	100	100	-100	-100	✓
100,000	1.357×10^{13}	1.357×10^{13}	-1.357×10^{13}	-1.357×10^{13}	✓
108,340	2.267×10^9	2.267×10^9	2.267×10^9	2.267×10^9	✓
108,360	2.274×10^9	2.274×10^9	2.274×10^9	2.274×10^9	✓

Table 1: Spot-check of the reconstructed liquidity map at block 23,479,243. All values match the on-chain `pool.ticks()` call exactly.

All ten ticks match the on-chain ground truth bit-for-bit, for both `liquidityGross` and `liquidityNet`.

Slot0 Consistency Check

At each of the 182 daily snapshots, we compare the price derived from `sqrtPriceX96` in `slot0_snapshots.parquet` to the price of the last recorded swap strictly before that block (`last_swap_price`). The column `price_diff_ticks` records the gap in tick units between the two prices. The median absolute discrepancy across all snapshots is **0 ticks**; the maximum observed is **2 ticks**, well below the pool's tick spacing of 10. There is no systematic drift, which confirms that `sqrtPriceX96` is decoded consistently with the swap event data.

Conclusion

The event replay reproduces the pool's tick-level state exactly at block 23,479,243. The extracted files are internally consistent and pass every on-chain spot-check. No discrepancies were found.

Module 2 — Liquidity Distribution Analysis

This module analyses how liquidity is distributed across price ranges in the USDC/WETH Uniswap V3 pool. Because Uniswap V3 liquidity is range-based, total TVL alone is not enough to describe market depth. The key question is whether liquidity is close to the current ETH price and whether it is concentrated in a small number of ticks.

Task 2.1 — Liquidity Profile Visualisation

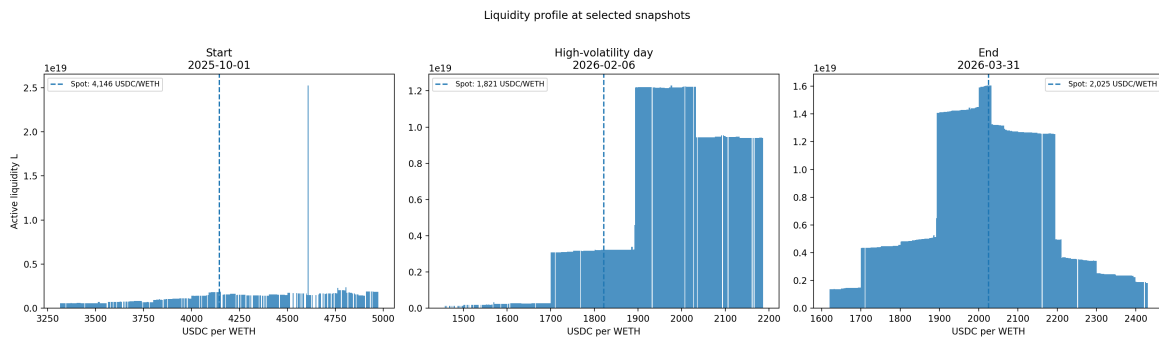


Figure 1: Liquidity Profile Visualisation

Fig. 1 shows liquidity profiles at three snapshots: the start of the sample, a high-volatility day, and the end of the sample. The dashed vertical line is the spot ETH price. At the start of the period, the spot price is around 4,146 USDC/WETH. Liquidity is spread over a wide range, with some large liquidity away from the current price. This means that not all liquidity is immediately useful for trades around spot. On the high-volatility day, the spot price falls to around 2,688 USDC/WETH. The profile becomes more uneven, with liquidity clustered in specific regions. This suggests that market depth varies strongly depending on the price interval: trades inside dense regions face lower slippage, while trades crossing sparse regions face higher slippage. By the end of the period, the spot price is around 2,025 USDC/WETH. Liquidity is much more concentrated around the current price, especially between roughly 1,900 and 2,200 USDC/WETH. This indicates that LPs adjusted to the lower ETH price regime and repositioned liquidity closer to the active trading range. The tiled liquidity profiles confirm this dynamic. Over time, liquidity migrates downward together with ETH price, moving from the 3,000–4,500 range toward the 1,900–2,300 range.

Task 2.2 — TVL Computation and Decomposition

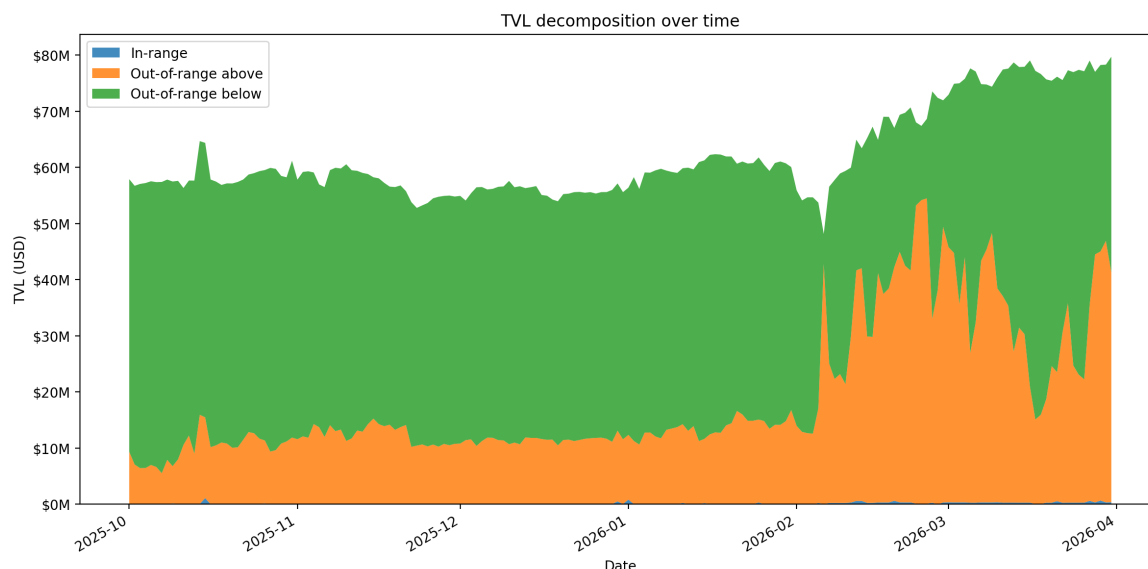


Figure 2: TVL Decomposition

Figure 2 decomposes TVL into in-range liquidity and out-of-range liquidity above and below the current price. The main result is that most TVL is out of range for most of the sample. Total TVL remains large, roughly between 50M and 80M USD, but the in-range component is very small. This means that the pool's headline TVL overstates the liquidity actually available for immediate trading. A sharp change occurs around February 2026, when ETH price falls strongly. The composition of out-of-range liquidity shifts, with a larger share becoming out-of-range above the current price. This reflects the market moving away from previously active LP ranges. Therefore, the pool has substantial locked capital, but much of it is inactive at the current spot price.

Task 2.3 — Liquidity Concentration Metrics

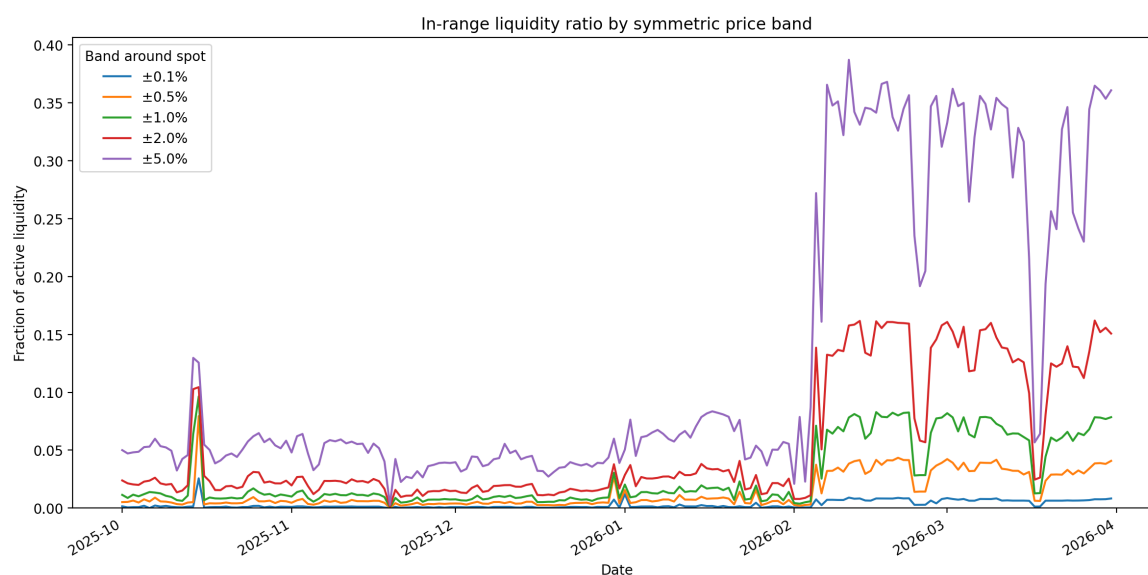


Figure 3: In-Range Liquidity Ratio

The In-Range Liquidity Ratio (ILR) measures the fraction of active liquidity located within a symmetric price band around the current pool price. Figure 3 reports ILR values for five bands: $\pm 0.1\%$, $\pm 0.5\%$, $\pm 1\%$, $\pm 2\%$, and $\pm 5\%$.

Before February 2026, liquidity close to the spot price is limited. The $\pm 5\%$ ILR generally remains below 10%, while the narrower bands are close to zero. This indicates that only a small fraction of total liquidity is positioned tightly around the current price.

After February 2026, ILR increases sharply. The $\pm 5\%$ band rises to approximately 30–38%, while the $\pm 2\%$ band increases to around 12–16%. This suggests that liquidity becomes substantially more concentrated near the spot price after ETH prices decline again. However, the $\pm 0.1\%$ band remains small, implying that liquidity is not concentrated extremely tightly at the exact spot price.

Overall, the ILR results show a significant improvement in near-price liquidity during the second half of the sample.

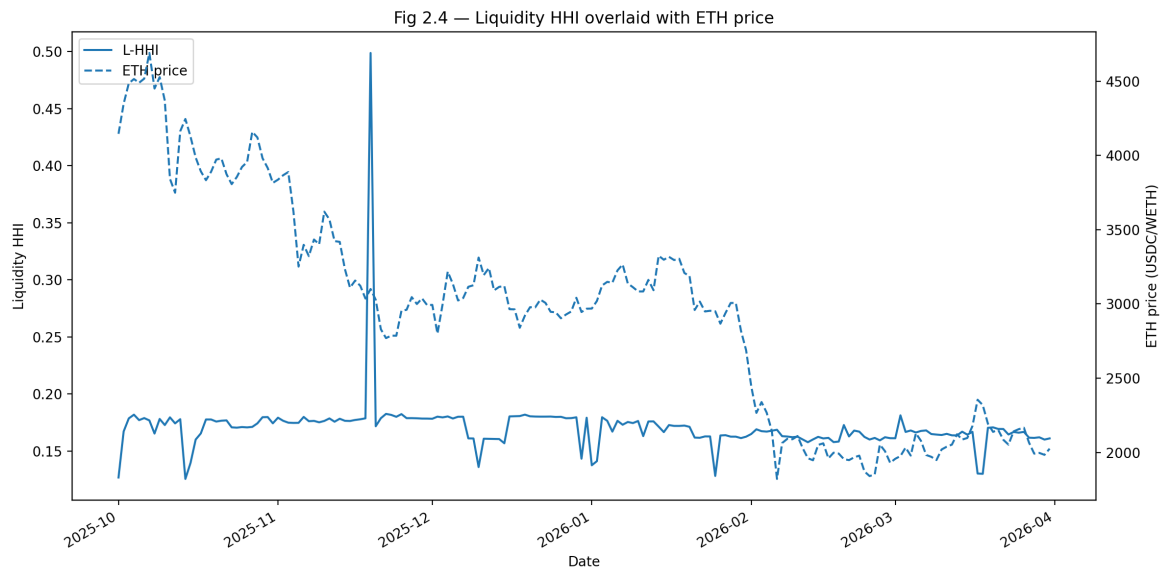


Figure 4: L-HHI

The L-HHI measures concentration across initialized ticks. A high value means liquidity is dominated by a few ticks, while a low value means liquidity is more evenly distributed. In the updated Figure 4, L-HHI is mostly between 0.15 and 0.18, with a slight downward trend toward the end of the sample. This means liquidity is moderately concentrated across ticks, but not fully dominated by one tick. The large temporary spike around late November indicates a short-lived concentration event, where liquidity became unusually concentrated in a small number of initialized ticks. The ETH price declines from above 4,000 USDC/WETH to around 2,000 USDC/WETH over the sample. Despite this large price decline, L-HHI remains relatively stable. This suggests that LPs changed where liquidity was placed, but liquidity concentration across ticks did not become permanently more extreme.

In conclusion, the pool adapts to the falling ETH price by moving liquidity toward lower price ranges. By March 2026, liquidity is more closely aligned with the current trading

region than at the start of the sample. However, most TVL remains out of range, so total TVL is not a reliable measure of usable market depth. The ILR metric shows that near-price liquidity improves after February, especially within wider bands such as $\pm 2\%$ and $\pm 5\%$. At the same time, the updated L-HHI shows only moderate tick-level concentration, meaning liquidity is not persistently controlled by a few isolated ticks.

Together, the metrics suggest that liquidity became more useful around the spot price after the ETH price decline, while remaining reasonably distributed across initialized ticks.

Module 3 — Slippage Simulation and Execution Cost

Module 3 quantifies trading costs in the USDC/WETH 0.05% pool by implementing a faithful tick-by-tick swap simulator, running it across a systematic grid of trade sizes and daily snapshots, and comparing the results to realised execution costs extracted from on-chain swap events.

Task 3.1 — Uniswap V3 Swap Simulator

3.1.1 Algorithm

The simulator reproduces the Uniswap V3 swap algorithm exactly. Given a reconstructed tick-level liquidity map $\{(t_i, \Delta L_i)\}$, the current pool state ($\text{sqrPriceX96}, \tau_{\text{cur}}, L_{\text{active}}$) from `slot0`, a direction flag (buy WETH or sell WETH), and a gross input amount, it iterates through tick intervals until the input is fully consumed.

At each step, the active liquidity L is constant within the interval $[\tau_{\text{lo}}, \tau_{\text{hi}}]$. Let s_c and s_t denote the current and target $\sqrt{P}$ values (continuous, not Q64.96-scaled). The per-step conservation laws are:

Buy WETH (`zeroForOne` = `True`, $\sqrt{P}$ decreasing):

$$\Delta x_{\text{gross}} = \frac{L}{1 - \phi} \left(\frac{1}{s_t} - \frac{1}{s_c} \right) / 10^6, \quad (1)$$

$$\Delta y_{\text{out}} = L (s_c - s_t) / 10^{18}, \quad (2)$$

Sell WETH (`zeroForOne` = `False`, $\sqrt{P}$ increasing):

$$\Delta y_{\text{gross}} = \frac{L}{1 - \phi} (s_t - s_c) / 10^{18}, \quad (3)$$

$$\Delta x_{\text{out}} = L \left(\frac{1}{s_c} - \frac{1}{s_t} \right) / 10^6, \quad (4)$$

where $\phi = 0.0005$ is the 0.05% fee rate and the $10^6/10^{18}$ factors convert USDC (6 decimals) and WETH (18 decimals) to human-readable units.

The fee is deducted from each gross input step before computing the output. At each tick interval, the loop first computes the maximum input that can be consumed before reaching the next initialised tick boundary. If the remaining input is insufficient to reach the boundary, the new $\sqrt{P}$ is solved analytically from Eqs. (1)–(4) and the loop terminates. Otherwise, the full step is consumed, the tick boundary is crossed, `liquidityNet` is applied to update L_{active} , and the loop continues to the next interval.

After the swap, price impact and net slippage are reported in basis points:

$$\text{impact (bps)} = \left(\frac{p_{\text{exec}}}{p_{\text{mid}}} - 1 \right) \times 10,000, \quad \text{slippage (bps)} = \text{impact} - \phi \times 10,000, \quad (5)$$

where $p_{\text{exec}} = \Delta x_{\text{in}} / \Delta y_{\text{out}}$ for a buy and p_{mid} is the pool price at the snapshot block.

Note: All `sqrPriceX96` values stored on-chain use Q64.96 fixed-point encoding. The simulator converts them to continuous floating-point $\sqrt{P}$ via $\sqrt{P} = \text{sqrPriceX96} / 2^{96}$ before any arithmetic, and applies the decimal offsets (10^6 for USDC, 10^{18} for WETH) at each step.

3.1.2 Validation

Twenty real swaps were drawn from `swap_events.parquet` with notional values between \$10 000 and \$500 000, spread uniformly across the study window. For each swap, the nearest prior daily snapshot was used as the starting pool state and the simulator was re-run with the observed gross input. The simulated output was then compared to the on-chain recorded output.

Swap Simulator Validation
n=20 Median |error| = 1.312%

Timestamp	Direction	Notional (\$)	Mid Price	Actual Out	Sim Out	Error (%)
2025-10-01 00:01	buy_weth	15,206	4,145.56	3.66464	3.66564	+0.027%
2025-10-10 23:51	buy_weth	10,315	4,369.08	2.66984	2.35951	-11.624%
2025-10-17 11:13	buy_weth	18,622	3,893.51	4.95493	4.77911	-3.548%
2025-10-30 02:09	buy_weth	33,032	3,915.62	8.40214	8.42887	+0.318%
2025-11-07 17:21	sell_weth	26,147	3,321.74	26146.58283	25975.79143	-0.653%
2025-11-16 16:38	buy_weth	15,131	3,157.30	4.92327	4.78915	-2.724%
2025-11-23 21:07	sell_weth	44,999	2,786.21	44999.37761	44291.15286	-1.574%
2025-12-08 14:38	sell_weth	41,307	3,115.16	41307.44645	41004.50475	-0.733%
2025-12-25 01:03	buy_weth	13,223	2,946.80	4.49674	4.48416	-0.280%
2026-01-15 15:27	buy_weth	100,943	3,297.60	30.26206	30.57987	+1.050%
2026-01-31 02:47	buy_weth	23,019	2,687.73	8.51244	8.55878	+0.544%
2026-02-05 14:50	buy_weth	118,654	2,143.61	57.10257	55.30991	-3.139%
2026-02-08 11:04	buy_weth	76,681	2,090.84	36.49836	36.65019	+0.416%
2026-02-13 16:30	buy_weth	46,037	1,946.88	22.39536	23.63293	+5.526%
2026-02-20 17:51	buy_weth	13,295	1,947.55	6.75817	6.82291	+0.958%
2026-02-27 13:23	buy_weth	35,577	2,007.92	18.16881	17.70861	-2.533%
2026-03-04 12:19	sell_weth	18,976	1,978.36	18976.30020	18363.29511	-3.230%
2026-03-10 16:25	buy_weth	445,339	2,019.49	213.46598	220.26925	+3.187%
2026-03-18 11:44	buy_weth	16,015	2,318.74	7.03440	6.90254	-1.875%
2026-03-25 00:48	sell_weth	16,240	2,156.36	16239.50962	16171.11467	-0.421%

Figure 5: Swap simulator validation on 20 real swaps. Green cells: $|\text{error}| < 1\%$; yellow: $< 5\%$; red: $\geq 5\%$.

The median absolute percentage error is **1.31%** and the mean is **2.22%** across the 20 sampled swaps (Fig. 5). Residual errors arise from two main sources: the snapshot used as the starting state is taken at the daily 00:00 UTC block, which may be up to 24 hours before the actual swap, so intra-day liquidity additions and removals are not captured; and for large trades that cross many ticks, small floating-point rounding differences in $\sqrt{P}$ accumulate across loop steps. Overall, the accuracy is sufficient for the price-impact analysis of Tasks 3.2–3.4.

Task 3.2 — Simulation Grid

3.2.1 Setup

The simulator was run across the full Cartesian grid of trade sizes, daily snapshots, and directions. Trade sizes span seven notional USD levels (\$1K, \$10K, \$50K, \$100K, \$250K, \$500K, and \$1M), applied to all 182 daily 00:00 UTC blocks from 1 October 2025 to 31 March 2026, in both the buy-WETH and sell-WETH directions. This yields $7 \times 182 \times 2 = 2,548$ simulated trades, saved as `simulated_trades.parquet`.

3.2.2 Results

Table 2 reports the distribution of simulated price impact for buy-WETH trades across the 182 daily snapshots.

Table 2: Simulated price impact (bps) for buy-WETH trades across 182 daily snapshots.

Notional	Median	Mean	Min	Max
\$1 000	5.10	5.10	5.01	5.42
\$10 000	5.96	6.01	5.05	9.18
\$50 000	9.79	10.02	5.23	25.83
\$100 000	14.57	15.02	5.46	46.61
\$250 000	28.92	30.10	6.15	108.49
\$500 000	53.40	55.26	7.30	209.78
\$1 000 000	102.03	104.39	9.61	407.11

At \$1K the median impact of 5.1 bps is almost entirely the 5 bps fee, with negligible market-depth impact. The min column shows that on the most liquid days even a \$500K trade incurs as little as 7.3 bps total impact, whereas the max column shows the same trade costing over 200 bps on thin days. The interquartile spread widens dramatically beyond \$50K, motivating the use of the full 182-snapshot panel.

Task 3.3 — Price Impact Curve Analysis

3.3.1 Log-Log Scaling

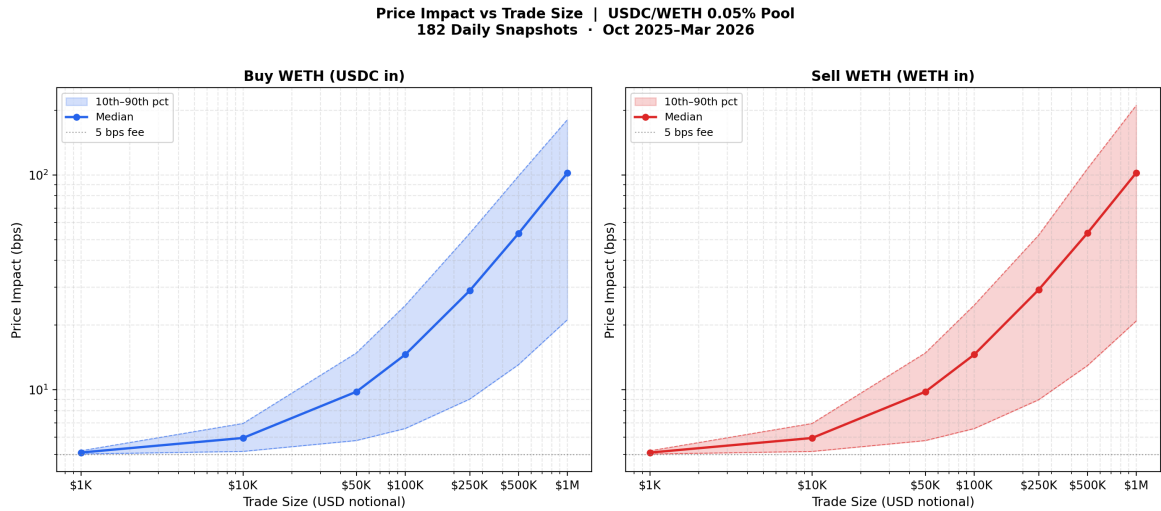


Figure 6: Price impact (bps) vs. trade size on a log-log scale. Median (solid line) and 10th–90th percentile band across 182 daily snapshots. Left: buy WETH; right: sell WETH.

Figure 6 shows a clear power-law relationship between trade size and price impact on the log-log scale, consistent with the square-root impact law observed in limit-order-book markets and predicted by AMM virtual-reserve geometry. For buy-WETH trades the median impact rises from 5.1 bps at \$1 000 to 102 bps at \$1 000 000, crossing the 5 bps fee threshold between the \$1K and \$10K buckets, and sell-WETH trades follow

a qualitatively similar profile. The 10th–90th percentile bands widen substantially at larger sizes, reflecting large day-to-day variability in in-range liquidity depth. The full range is even wider: across the 182 snapshots, the best-case impact for a 250K trade is as low as 6.2 bps while the worst-case exceeds 108 bps.

3.3.2 Power-Law Regression

To quantify the scaling relationship, we fit the model

$$\text{impact (bps)} = a \times Q^b, \quad (6)$$

where Q is the notional trade size in USD, by ordinary least squares on the log-transformed data

$$\log(\text{impact}) = \log a + b \log Q. \quad (7)$$

Table 3 reports the estimated coefficients and goodness of fit.

Table 3: Power-law regression of price impact on trade size. Fit: $\text{impact (bps)} = a \times Q^b$, estimated by log-log OLS across all 182 daily snapshots.

Direction	a (scale)	b (exponent)	R^2
Buy WETH	0.2115	0.3905	0.6266
Sell WETH	0.2088	0.3919	0.6172

The estimated exponents are $b = 0.3905$ (buy) and $b = 0.3919$ (sell), both meaningfully below the theoretical square-root value of 0.5 predicted by AMM virtual-reserve geometry. This sub-square-root scaling suggests that the in-range liquidity of this pool deepens faster than linearly with distance from the current price, so larger trades encounter progressively more depth rather than a uniform pool. The scale coefficients are nearly identical across directions ($a = 0.2115$ vs. 0.2088), confirming the near-symmetry between buy and sell impact at equivalent notional sizes. The R^2 of 0.63 (buy) and 0.62 (sell) indicates that a single power law captures roughly two-thirds of the cross-sectional variation; the remaining third reflects day-to-day fluctuations in in-range liquidity depth that a static exponent cannot capture, consistent with the wide percentile bands seen in Fig 6. The regression is fitted on all $7 \times 182 = 1,274$ observations per direction jointly; results are saved to `price_impact_regression.parquet`.

Task 3.4 — Effective Spread from Observed Swaps

3.4.1 Method

The realised effective spread is estimated directly from `swap_events.parquet` without simulation, using the standard microstructure formula:

$$\text{ES}_i = 2 D_i \frac{p_i^{\text{exec}} - p_i^{\text{mid}}}{p_i^{\text{mid}}}, \quad D_i = +1 \text{ (buy)}, -1 \text{ (sell)}, \quad (8)$$

where $p_i^{\text{exec}} = |\Delta x|/|\Delta y|$ is the trade-implied execution price in USDC/WETH and p_i^{mid} is the exact pool price at the nearest prior daily snapshot block. This calculation was applied to all 1,058,506 swaps in the study window.

3.4.2 Results

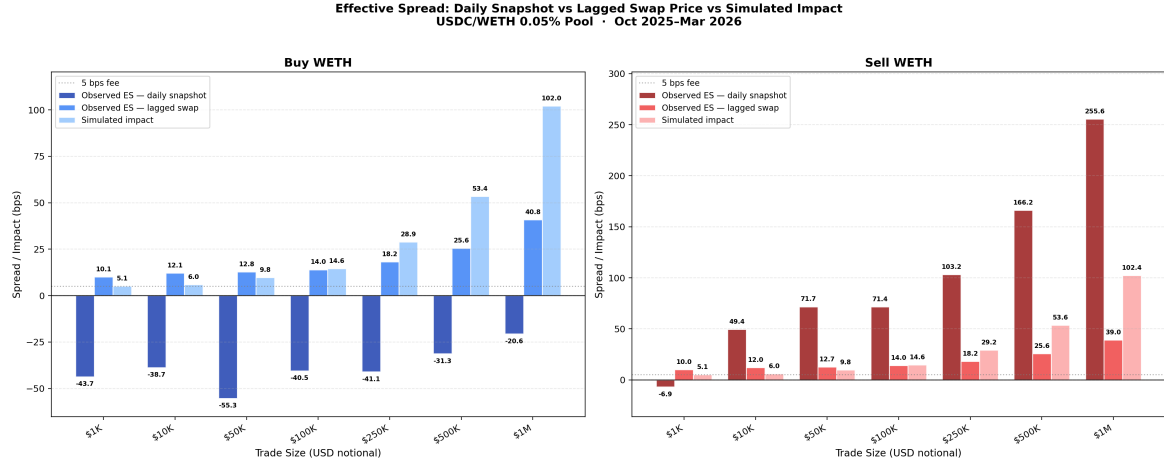


Figure 7: Median effective spread under two mid-price conventions (daily snapshot: dark bars; lagged swap price: mid-tone bars) vs. median simulated impact (light bars), by trade-size bucket. Left: buy WETH; right: sell WETH.

Sell WETH. The daily-snapshot effective spread (dark bars) starts at -6.91 bps at $\$1K$ and grows to 255.6 bps at $\$1M$, with a sign reversal between those extremes. At small sizes the stale mid-price dominates: the 24-hour drift pushes the snapshot price below the actual trade price, mechanically deflating the measured spread below zero. At large sizes the price impact term overwhelms the drift, producing large positive values that substantially exceed the simulated impact (255.6 vs. 102.4 bps at $\$1M$), inflating the observed spread above the true execution cost. The lagged spread (mid-tone bars), by contrast, is uniformly positive and modest, rising from 10.0 bps at $\$1K$ to 39.0 bps at $\$1M$, and tracks the simulated impact (5.1 to 102.4 bps) closely at small sizes while remaining below it at large sizes.

Buy WETH. The daily-snapshot effective spread is consistently *negative* across all size buckets (ranging from -43.7 bps at $\$1K$ to -20.6 bps at $\$1M$), the opposite sign of the strictly positive simulated impact. During a period of sustained ETH appreciation the snapshot price systematically understates the price prevailing at trade time, yielding a mechanically negative ES regardless of trade size, with a magnitude consistent with the 24-hour intra-day ETH price drift. The lagged spread removes this artefact entirely: it is uniformly positive, rising from 10.1 bps at $\$1K$ to 40.8 bps at $\$1M$, and aligns closely with the simulated impact across all buckets.

Supplementary extension (lagged mid-price correction). Although the instructions require p^{mid} to be taken from the nearest daily `slot0` snapshot, we implement here a zero-cost improvement that more closely approximates the pool price at the exact block before each swap. Because each Swap event records the pool price after the swap, the price prevailing immediately before swap i is simply the `sqrtPriceX96` of swap $i - 1$, available directly in `swap_events.parquet` without any additional RPC calls. Replacing the daily snapshot with this lagged price eliminates the 24-hour drift bias in both directions simultaneously. The close agreement between the lagged observed spread and the simulated impact across all seven size buckets and both trade

directions provides independent validation of the swap simulator: the tick-by-tick model correctly predicts the true execution cost once the measurement bias is removed.

Module 4: Liquidity Provision Analytics

This module analyses the Uniswap V3 USDC/WETH 0.05% pool (contract `0x88e6A0c2dDD26FEEb64F039a2c41296FcB3f5640`) from the perspective of a passive liquidity provider. Five synthetic positions of \$100,000 notional are entered at the first daily snapshot (1 October 2025, $p_0 = \$4,145.56$ USDC/WETH) and held until the last (31 March 2026, $p_T = \$2,024.66$), representing a -51.2% adversarial price movement over the six-month study window.

D4.1 — Methodology and Position Construction

Price Derivation from `sqrtPriceX96`

All prices are recomputed directly from the `sqrtPriceX96` field stored in `slot0_snapshots.parquet`, following the exact Uniswap V3 on-chain encoding:

$$P_{\text{raw}} = \left(\frac{\text{sqrtPriceX96}}{2^{96}} \right)^2 = \frac{\text{WETH}_{\text{raw}}}{\text{USDC}_{\text{raw}}} \quad (9)$$

$$p [\text{USDC/WETH}] = \frac{10^{12}}{P_{\text{raw}}} \quad (10)$$

The factor 10^{12} corrects for the decimal asymmetry between USDC (6 decimals) and WETH (18 decimals): $10^{18}/10^6 = 10^{12}$.

Virtual Reserve Formulas

The USD mark-to-market value of a position with liquidity L , current sqrt-price $\sqrt{p_c}$, and range boundaries $\sqrt{p_a}$ (lower tick) and $\sqrt{p_b}$ (upper tick) is derived from the Uniswap V3 whitepaper three-case formula:

$$(x, y) = \begin{cases} \left(\frac{L(\sqrt{p_b} - \sqrt{p_a})}{\sqrt{p_a}\sqrt{p_b}}, 0 \right) & \sqrt{p_c} \leq \sqrt{p_a} \\ \left(\frac{L(\sqrt{p_b} - \sqrt{p_c})}{\sqrt{p_c}\sqrt{p_b}}, L(\sqrt{p_c} - \sqrt{p_a}) \right) & \sqrt{p_a} < \sqrt{p_c} < \sqrt{p_b} \\ (0, L(\sqrt{p_b} - \sqrt{p_a})) & \sqrt{p_c} \geq \sqrt{p_b} \end{cases} \quad (11)$$

where x is USDC in raw units and y is WETH in raw units. The USD value is $V_{LP} = x/10^6 + (y/10^{18}) \cdot p$. Since Eq. (11) is linear in L , the required liquidity for a \$100,000 notional is:

$$L = \frac{\$100,000}{V_{LP}(L=1)} \quad (12)$$

Tick Spacing Compliance

The 0.05% pool enforces a tick spacing of 10: every initialised tick must be an exact multiple of 10. Continuous tick values derived from the target price boundaries $[p_a, p_b]$ are snapped to the nearest valid grid points using outward rounding:

$$\text{tick_lower} = \lfloor t_a/10 \rfloor \times 10 \quad (13)$$

$$\text{tick_upper} = (\lceil t_b/10 \rceil + 1) \times 10 \quad (14)$$

This guarantees that the entry price sits strictly inside the range $[\text{tick_lower}, \text{tick_upper})$ as required by the protocol.

Position Definitions and Non-Compounding Assumption

The five positions studied are defined in Table 4. Throughout the simulation, the liquidity constant L is held *fixed* at its entry value (non-compounding assumption): fees are accumulated without reinvestment, reflecting the on-chain reality that compounding requires an explicit `collect()` transaction followed by re-minting.

Table 4: LP Position Parameters at Entry (1 Oct 2025, $p_0 = \$4,145.56$)

Pos	Band	p_a [\$/ETH]	p_b [\$/ETH]
P1	$\pm 0.1\%$	4138.1	4150.5
P2	$\pm 0.5\%$	4121.6	4167.1
P3	$\pm 2\%$	4060.2	4231.1
P4	$\pm 10\%$	3729.4	4572.0
P5	Full range	≈ 0	∞

D4.2 — Fee Income Computation

For each swap where the active tick falls inside $[\text{tick_lower}, \text{tick_upper})$, the LP earns a pro-rata share of the pool’s fee revenue:

$$\text{LP fee}_t = \frac{L_{\text{position}}}{L_{\text{active},t}} \times \text{pool fee}_t \quad (15)$$

The pool fee per swap is $0.05\% \times |\Delta\text{USDC}|$ for buy-WETH swaps and $0.05\% \times |\Delta\text{WETH}| \times p_t$ for sell-WETH swaps. The ratio $L_{\text{position}}/L_{\text{active},t}$ in Eq. (15) reflects the pro-rata mechanics of Uniswap V3: a position with higher concentrated liquidity captures a larger fee share *per in-range swap*, at the cost of a narrower active price window.

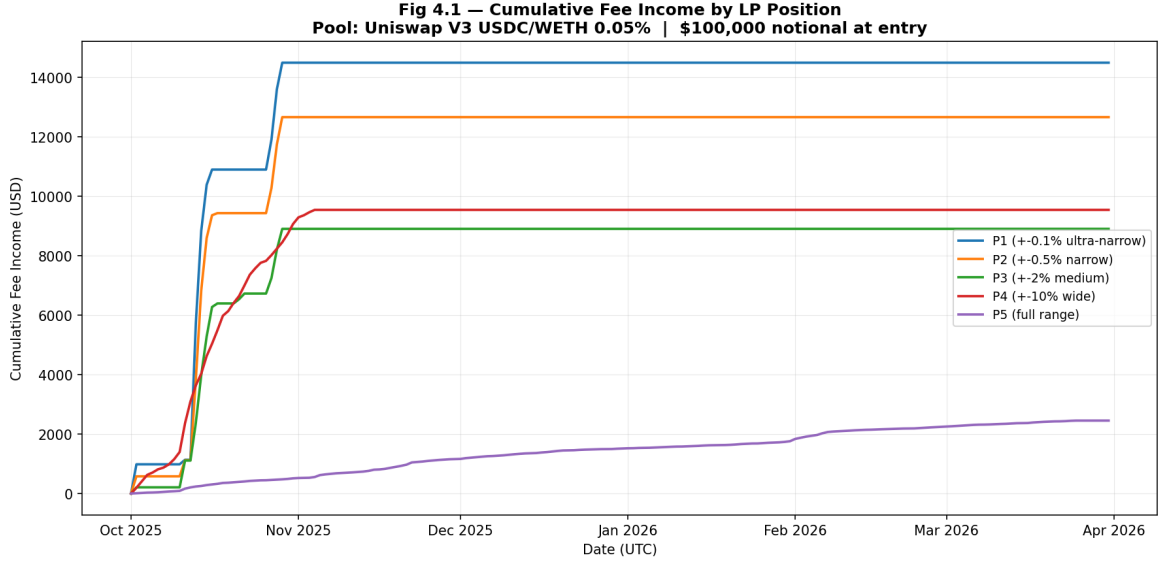


Figure 8: Cumulative fee income by LP position (Oct 2025 – Mar 2026). Flat plateaus correspond to periods when the pool price lies outside the position range and the position earns zero fees.

Table 5: In-Range Activity Statistics per Position

Pos	Band	% of days in range	First exit
P1	$\pm 0.1\%$	0.5%	2 Oct 2025
P2	$\pm 0.5\%$	2.2%	2 Oct 2025
P3	$\pm 2\%$	2.7%	2 Oct 2025
P4	$\pm 10\%$	18.1%	7 Oct 2025
P5	Full range	100%	Never

As shown in Fig. 8 and Table 5, narrower positions earn fees at a higher rate per active swap due to the leverage effect of concentrated liquidity, but this advantage is severely curtailed by the brevity of their in-range periods. P1 exits its range on the *second day* of the study window and is active for only 0.5% of daily snapshots. This illustrates the fundamental fragility of ultra-concentrated positions in trending markets: without continuous rebalancing, the leverage effect becomes a liability rather than an asset.

D4.3 — Impermanent Loss and Net P&L

Impermanent loss at time t is defined as the opportunity cost of providing liquidity versus holding the initial portfolio:

$$IL(t) = V_{\text{HODL}}(t) - V_{LP}(t), \quad V_{\text{HODL}}(t) = x_0 \cdot p_t + y_0 \quad (16)$$

where x_0 and y_0 are the WETH and USDC balances at entry computed from Eq. (11). Net P&L combines fee income and impermanent loss:

$$\text{Net P\&L}(t) = \sum_{\tau \leq t} \text{LP fee}_{\tau} - IL(t) \quad (17)$$

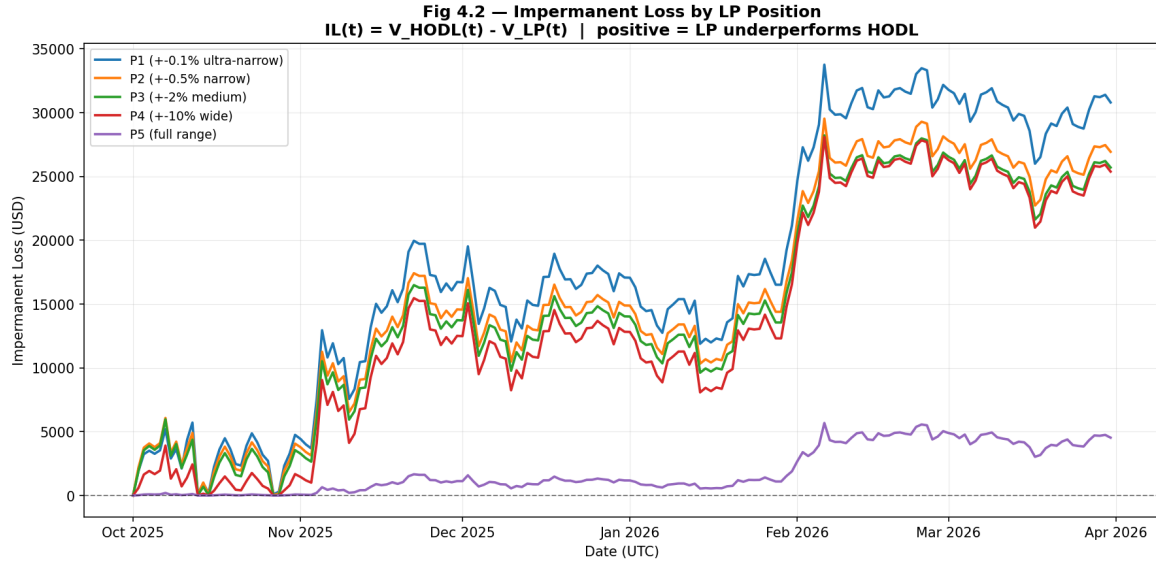


Figure 9: Impermanent loss by LP position. $IL(t) > 0$ means the LP underperforms a passive HODL strategy. The rapid divergence in early October 2025 corresponds to bounded positions exiting their active ranges as ETH declines.

Table 6: Terminal Results (31 March 2026, $p_T = \$2,024.66$, entry $p_0 = \$4,145.56$, change = -51.2%)

Pos	Fees	Terminal IL	Net P&L
P1 ($\pm 0.1\%$)	\$14,494	\$30,800	-\$16,307
P2 ($\pm 0.5\%$)	\$12,662	\$26,935	-\$14,272
P3 ($\pm 2\%$)	\$8,904	\$25,703	-\$16,799
P4 ($\pm 10\%$)	\$9,543	\$25,379	-\$15,837
P5 (Full range)	\$2,454	\$4,535	-\$2,081

Fig. 9 and Table 6 show that all bounded positions (P1–P4) transition to 100% WETH exposure within the first week of October 2025, after which V_{LP} falls in lockstep with V_{HODL} and impermanent loss grows monotonically with the price decline.

D4.4 — Analysis and Interpretation

The Paradox of Concentrated Liquidity Under Adverse Drift

Table 6 reveals a counter-intuitive result: P1, the most concentrated position, earns the highest absolute fee income (\$14,494) yet produces the second-worst net P&L (−\$16,307). Two effects compound here. First, the leverage effect of concentrated liquidity amplifies both fee capture and impermanent loss symmetrically: P1’s L is approximately $3.7\times$ larger than P2’s for the same \$100,000 notional, proportionally magnifying all mark-to-market fluctuations. Second, the extreme path-dependency of a $\pm 0.1\%$ position means that the -51.2% price decline renders it inactive within 24 hours of entry (first exit: 2 October 2025). Thereafter, P1’s portfolio is entirely WETH, accumulating zero additional fees while growing its impermanent loss in lockstep with the falling ETH price.

P5 as a Structural Benchmark

The full-range position P5 stands apart with a net P&L of $-\$2,081$, roughly $8\times$ less severe than the concentrated positions, despite earning the lowest fee income ($\$2,454$). P5 benefits from structural resilience: unlike bounded positions which become one-sided bets once the price exits the range, P5 always maintains exposure to both USDC and WETH, limiting its impermanent loss to $\$4,535$ — a factor of $5.6\times$ lower than P1. P5 therefore provides the most robust risk-adjusted outcome under adversarial price movement, at the cost of minimal fee efficiency.

High-Volatility Periods and the Discontinuity of Impermanent Loss

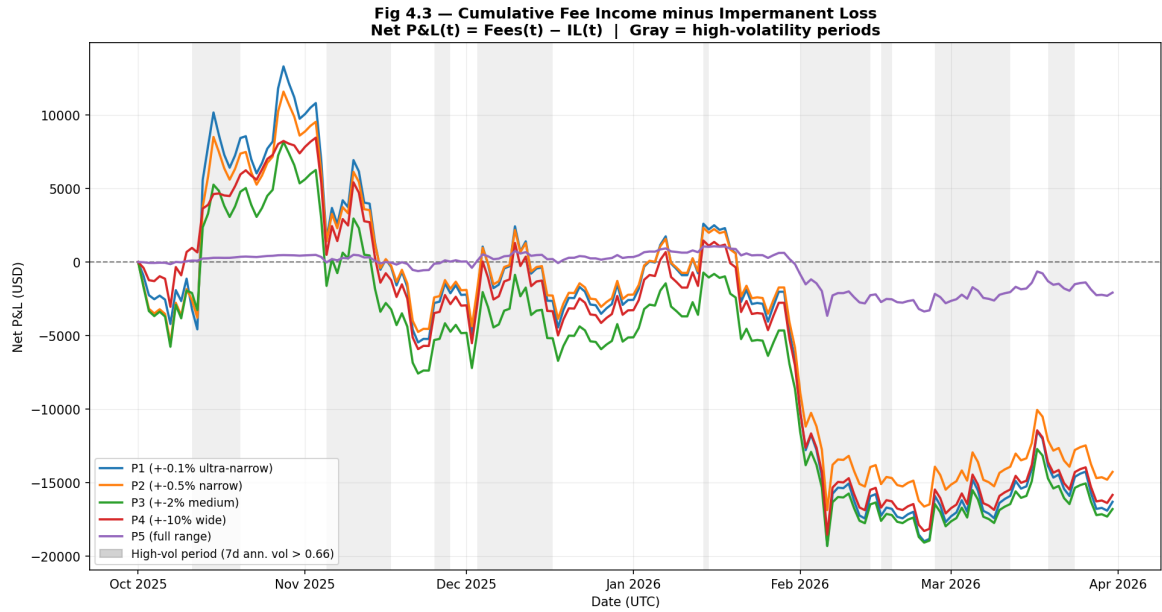


Figure 10: Net P&L with high-volatility bands (gray shading: 7-day rolling annualised volatility > 0.662 , the 60th percentile of the full window, yielding 9 distinct bands). Every sharp downward spike in net P&L coincides with a high-volatility episode.

Fig. 10 plots net P&L with gray shading identifying high-volatility periods, defined as days where the 7-day rolling realised annualised volatility exceeds its 60th percentile (threshold $\hat{\sigma} = 0.662$). A key structural insight emerges: fee income accumulates *continuously* and smoothly, proportional to swap volume, while impermanent loss is a *convex, path-dependent* function of price that can jump sharply during volatile episodes. During the nine high-volatility bands identified in Fig. 10, impermanent loss absorbs months of accumulated fee income in a matter of days. This asymmetry — slow fee accrual versus fast IL realisation — is the fundamental economic risk of passive liquidity provision under directional price trends.

Transition to Module 5: The Case for Delta-Hedging

The uniformly negative net P&L across all five positions (Table 6) is not merely a consequence of an unfavourable price path. It constitutes a quantitative proof that passive LP strategies carry an unhedged directional risk that fees alone cannot compensate in trending markets. As derived from Eq. (11), a Uniswap V3 LP position

is economically equivalent to a *short gamma* instrument: it profits from range-bound price action (fee income) but suffers a convex loss from large price moves (impermanent loss). This creates a delta exposure that grows monotonically with $|p_t - p_0|$ and cannot be diversified away within the LP framework alone. Module 5 addresses precisely this risk by constructing a derivatives overlay — perpetual futures or options-based delta-hedging — that neutralises the directional component of the LP position and isolates the fee income as a near-pure volatility carry. The results of Module 4 thus provide the scientific motivation for the hedging strategies developed next.

Module 5 — Dynamic Hedging of Impermanent Loss

Task 5.1 — Theoretical Framework

5.1.1 LP Payoff Shape and the Short Gamma Interpretation

To analyze the LP payoff, we evaluate the position value $V_{LP}(p)$ as a function of the terminal ETH price p . For a position bounded by $[p_a, p_b]$ with a liquidity constant L , we have seen during the lecture that the LP value is defined by:

$$V_{LP}(p) = \begin{cases} L \left(\frac{1}{\sqrt{p_a}} - \frac{1}{\sqrt{p_b}} \right) p & \text{if } p < p_a \\ L \left(2\sqrt{p} - \frac{p}{\sqrt{p_b}} - \sqrt{p_a} \right) & \text{if } p \in [p_a, p_b] \\ L (\sqrt{p_b} - \sqrt{p_a}) & \text{if } p > p_b \end{cases} \quad (18)$$

Note: The liquidity L is a constant representing the "depth" of the specific position, not the global pool liquidity (as we initially thought...). Since the initial notional is fixed at $V_0 = \$100,000$, L dynamically adjusts to compensate for the range width:

$$L = \frac{V_0}{2\sqrt{p_0} - \frac{p_0}{\sqrt{p_b}} - \sqrt{p_a}} \quad (19)$$

Comment on the payoff shape and its similarity to well-known option payoffs:

1. Payoff shape:
 - (a) We can see in the first plot of 12 that P1 case makes the transition from 100% ETH to 100% USDC almost instantly. The value of L is huge since the liquidity has to be provided in a very small area, then gamma has to be big, which is the case when L is large (see 5.1.3 section);
 - (b) For the others cases (ie P2, P3, P4 and P5), the shape is quite the same, with a larger zone in the middle, corresponding to a lower L value.
2. Financial parallel : We've seen in the lecture that these situations are *short put* positions but with something else in the interval $[p_a, p_b]$. A representation of this classical option payoffs is given in figure 11.

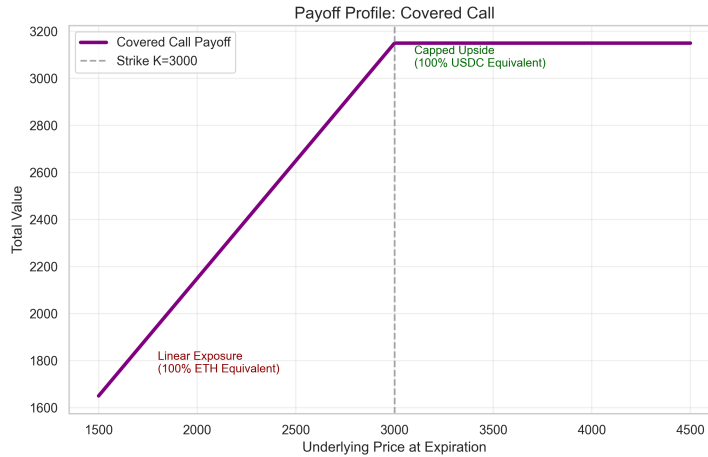


Figure 11: Payoff Profile: Covered Call

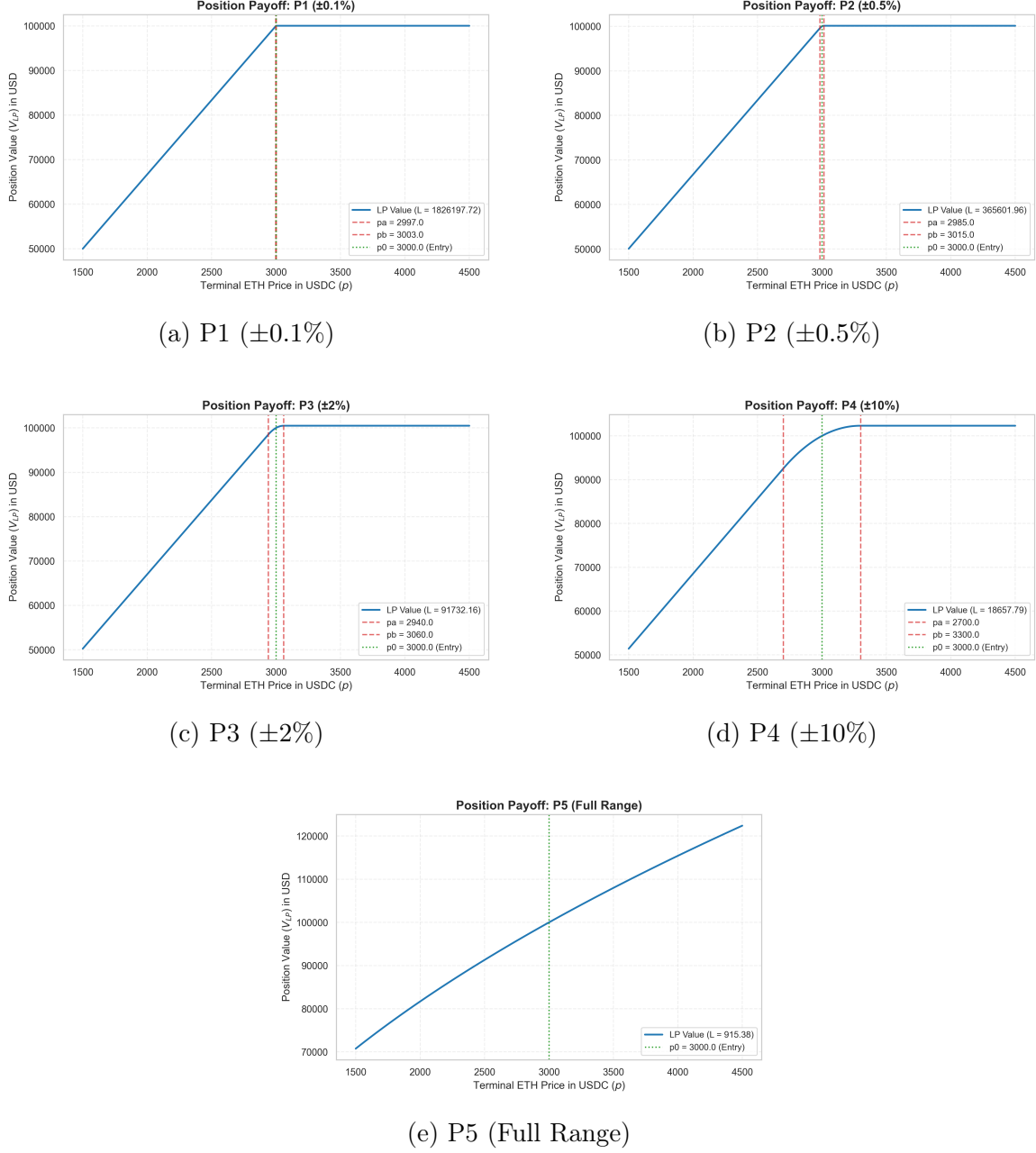


Figure 12: Payoff profiles for the 5 representative positions (P1 to P5). The active concave zone widens as the price range increases, while L decreases.

5.1.2 Derivation of LP Delta

By definition, the Delta (Δ_{LP}) of the LP position is the first derivative of V_{LP} with respect to the ETH price p . By differentiating the function from 5.1.1 (here 18), we obtain:

$$\Delta_{LP} = \frac{\partial V_{LP}}{\partial p} = \begin{cases} L \left(\frac{1}{\sqrt{p_a}} - \frac{1}{\sqrt{p_b}} \right) & \text{if } p < p_a \\ L \left(\frac{1}{\sqrt{p}} - \frac{1}{\sqrt{p_b}} \right) & \text{if } p \in [p_a, p_b] \\ 0 & \text{if } p > p_b \end{cases} \quad (20)$$

Important note: Mathematically, this derivative exactly matches the formula for the ETH inventory $x(p)$ held in the Uniswap V3 pool.

$$\Delta_{LP}(p) = \frac{\partial V_{LP}(p)}{\partial p} = \frac{\partial}{\partial p}[x(p) * p + y(p)] = x + p \frac{\partial x(p)}{\partial p} + \frac{\partial y(p)}{\partial p} = x(p) \quad (21)$$

since the internal mechanics of the AMM dictate that $dy = -p \cdot dx$

This confirms the financial intuition: directional risk stems exclusively from the physical quantity of the volatile asset held. Thus, $\Delta_{LP}(p) = x(p)$.

5.1.3 Gamma

Gamma (Γ_{LP}) is the second derivative of the position value with respect to price. Differentiating Δ_{LP} in the active range yields:

$$\Gamma_{LP} = \frac{\partial^2}{\partial p^2}[V_{LP}(p)] = \begin{cases} -\frac{L}{2p^{3/2}} & \text{if } p \in [p_a, p_b] \\ 0 & \text{if } p \notin [p_a, p_b] \end{cases} \quad (22)$$

The sign and magnitude of gamma link with LP position width:

1. The negative sign mathematically proves the **short gamma** nature of the LP position (meaning Δ_{LP} (then $x(p)$) is decreasing when p is increasing);
2. The magnitude of Gamma is directly proportional to L . Consequently, ultra-narrow ranges (like P1) require a massive L to reach the \$100,000 value, resulting in extreme Gamma risk over a very localized area.

Task 5.2 — Data Collection from Hyperliquid

5.2.1 Perp Price Data

Hourly OHLCV candles for the ETH perpetual contract were retrieved via the [Hyperliquid API](#).

Note 1: As indicates [here](#), we had to use milliseconds in order to create the json taken by the POST request

Note 2: Over a 6-month study window, rendering full candlesticks yields an illegible chart. Therefore, only the hourly close prices were plotted on figure 13 to represent the underlying market trajectory, which will serve as the mark price for the hedge P&L.

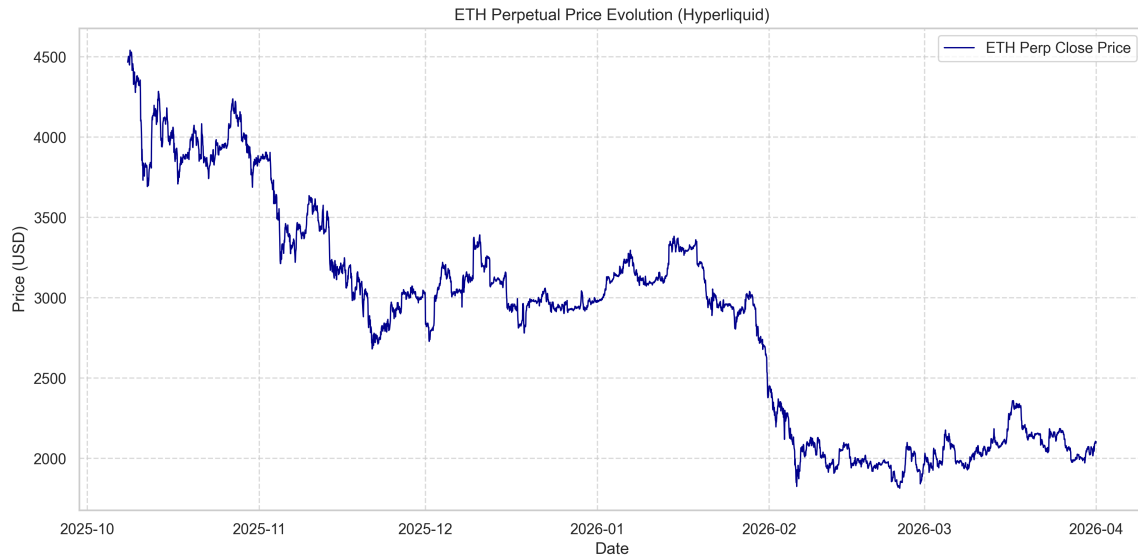


Figure 13: ETH Perpetual Close Price over the 6-month study window

5.2.2 Funding Rate Data

Data Extraction Challenge: While the Hyperliquid API documentation does not explicitly state a pagination limit for the `fundingHistory` endpoint, empirical testing revealed a hard cap of 500 records per request. To ensure complete and contiguous data extraction over 6 months, a timestamp-based pagination loop was engineered. Funding rates were extracted to accurately account for the holding cost/yield of the short hedge.

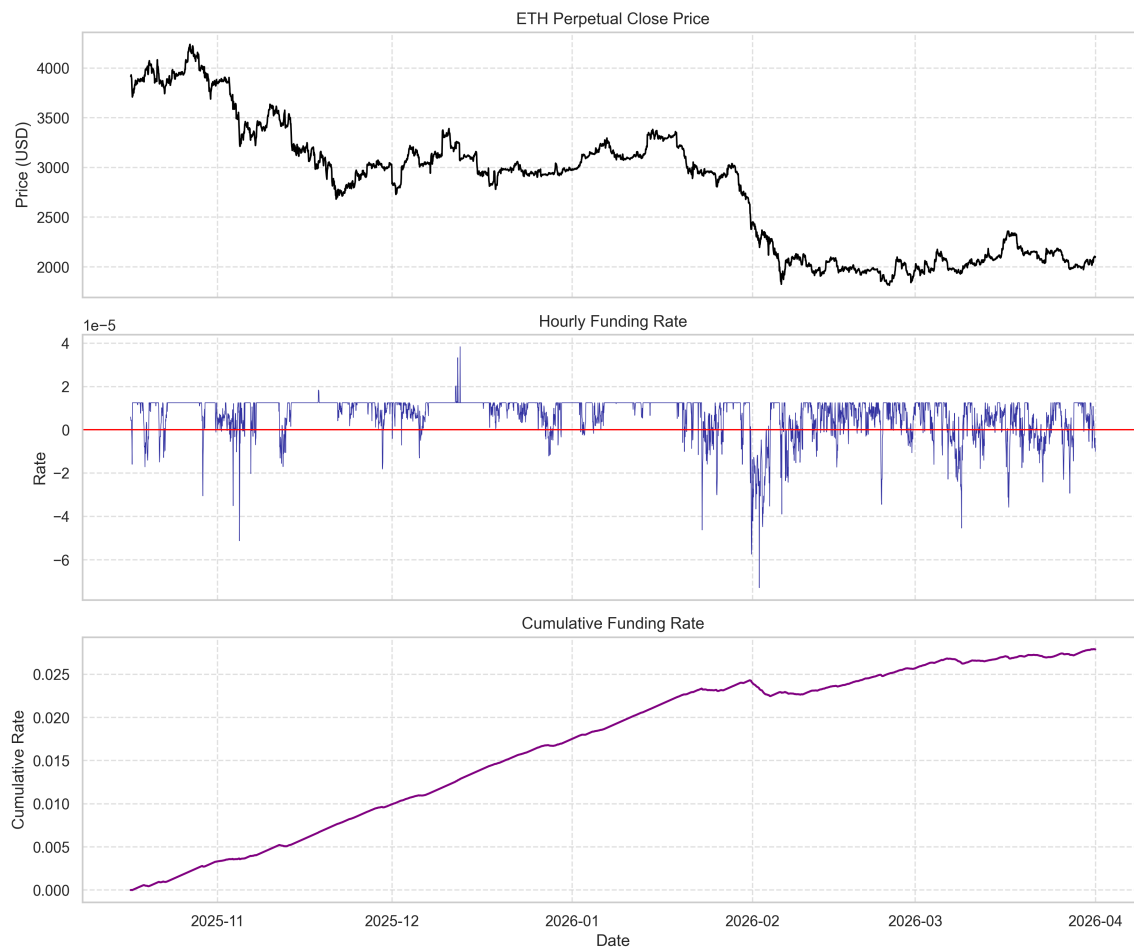


Figure 14: Hourly funding rate and cumulative funding rate over the study window

The data retrieved from Hyperliquid provides critical insights into the cost of maintaining a hedge during the study window. Two primary observations emerge from Figure 14:

1. **The Upward Drift of Cumulative Funding:** At first glance, the predominantly positive and increasing slope of the Cumulative Funding Rate might seem counter-intuitive given that the ETH spot price is trending downward over the same period. In a traditional directional market, one might expect negative funding during a bear trend (shorts paying longs). However, the results show that for the vast majority of the window, the funding rate remained positive.

This indicates that despite the overall price drop, market participants maintained a structural demand for long leverage on perps, causing the perp price to frequently trade at a slight premium relative to the oracle price. For a short hedger, this is a **favorable tailwind**: systematically *receiving* these funding payments helps offset some of the Impermanent Loss costs.

2. **Negative Spikes During Market Capitulation:** Rather than a single early flash crash, the hourly funding rate exhibits localized negative spikes that closely correlate with sharp downward price movements. The most extreme cluster occurs around late January and early February 2026, where the rate plunges to

approximately -7×10^{-5} . This coincides directly with the steepest price drop of the period (ETH falling rapidly from $\sim \$3,000$ to $\sim \$2,000$). This behavior reflects a localized panic and a sudden surge in demand for short positions, which temporarily inverted the funding rate and caused a visible dip in the otherwise upward trajectory of the Cumulative Funding curve.

Additional analysis

I wanted to analyse the chart during a small period of time to see the theory of funding rate.

I just asked Gemini to give me a code that emphasizes the zones where the market is bullish or bearish in order to see if the response concerning funding rate is logical.

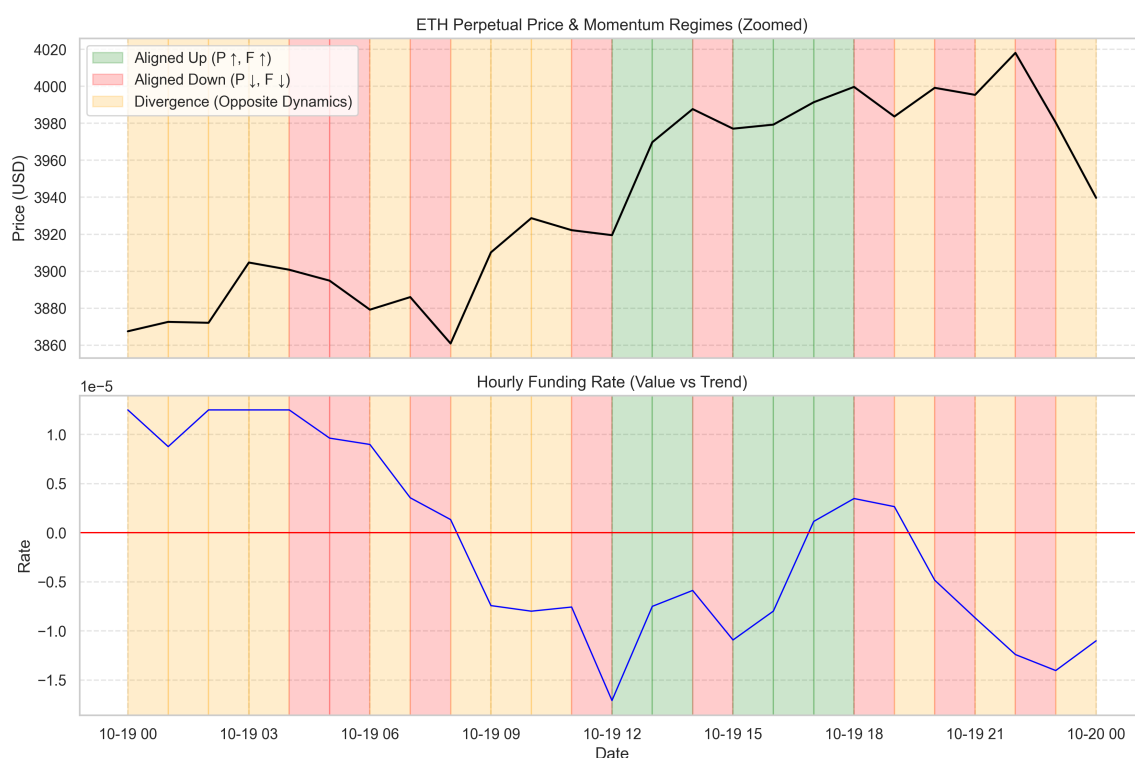


Figure 15: Hourly funding rate and cumulative funding rate - zoomed analysis

Analysis :

This graph is particularly interesting as it clearly illustrates the various market scenarios:

1. **RED COLOR:** The price of ETH falls: the market is bearish, prompting traders to short the market. As we saw earlier, the funding rate decreases (and tends to turn negative) because there is an excess of short positions, meaning the futures price falls below the spot price;
2. **GREEN COLOR:** Conversely, the market becomes bullish: traders take long positions, leading to an excess of buyers. Consequently, the funding rate must be positive, and it does indeed rise;

3. **ORANGE COLOR:** There are, however, special cases: such as the divergence shown in orange, where between 10-19 18 and 10-19 21, the price rises but the funding rate becomes negative. This can be explained by a spot price that rises very quickly, faster than the futures price. Because the perpetual contract lags behind and trades at a discount to the spot market, the funding rate turns negative.

Task 5.3 — Delta Hedging Backtest

The backtesting engine was designed to implement a discrete delta-hedging strategy. To maintain a Delta-Neutral portfolio against Impermanent Loss, the hedge target is the current ETH inventory of the LP position, $x(p) = L \left(\frac{1}{\sqrt{p}} - \frac{1}{\sqrt{p_b}} \right)$, which equals the LP delta $\frac{\partial V_{LP}}{\partial p}$. Shorting this exact quantity on the perpetual market neutralizes the directional ETH exposure of the position.

Here are the main difficulties encountered during this part :

- **Timestamp Misalignment in `df_merged`:** Funding rate timestamps contained sub-second offsets (e.g. 00:00:00.098), preventing exact joins with hourly candle data and reducing `df_merged` to only 57 rows.

Fix: Both series were floored to the hour via `.dt.floor('h')` before merging, restoring $\sim 4,200$ rows.

- **P_0 Misalignment:** LP positions from Module 4 were anchored at $\sim \$4,100$ (first `slot0` snapshot of the study window), while the corrected `df_merged` starts at $\sim \$4,524$ because the inner join drops candles predating the earliest available Hyperliquid funding rate record. This placed P1–P3 immediately out-of-range at $t = 0$ (see the notebook for the proof), freezing their gross IL at its maximum and producing zero hedge delta from the first timestamp.

Fix: Positions were rebuilt using $P_0 = \text{df_merged}['close'].iloc[0]$, with bounds $[p(1 - \delta), p(1 + \delta)]$ and L recomputed in human-readable units via $L = V_0 / (2\sqrt{p_0} - p_0 / \sqrt{p_b} - \sqrt{p_a})$.

- **L Unit Mismatch:** Raw L values from Module 4 incorporate token decimals (10^{18} for WETH, 10^6 for USDC), yielding $L \sim 10^{16}$ in on-chain units. Plugging this directly into human-readable USD formulas such as $V_{LP} = L(2\sqrt{p} - p/\sqrt{p_b} - \sqrt{p_a})$ produces values on the order of 10^{17} USD and integer overflow (`INT32_MIN` = $-2,147,483,648$) in aggregated metrics.

Fix: L is solved directly in USD-consistent units by normalising token amounts ($a_0/10^6$ for USDC, $a_1/10^{18}$ for WETH) before inverting the value equation, making it compatible with all downstream `lp_delta` and `lp_value` calls.

- **Hedge Target Definition:** An initial implementation shorted the full LP delta $x(p)$, which neutralises the LP's entire ETH exposure rather than only the IL-generating component. Over a period where ETH fell $\sim 50\%$, this produced a net hedge P&L of $\sim \$50,000$ against a gross IL of $\sim \$25,000$.

This behaviour is discussed further in Section 5.4.

Results

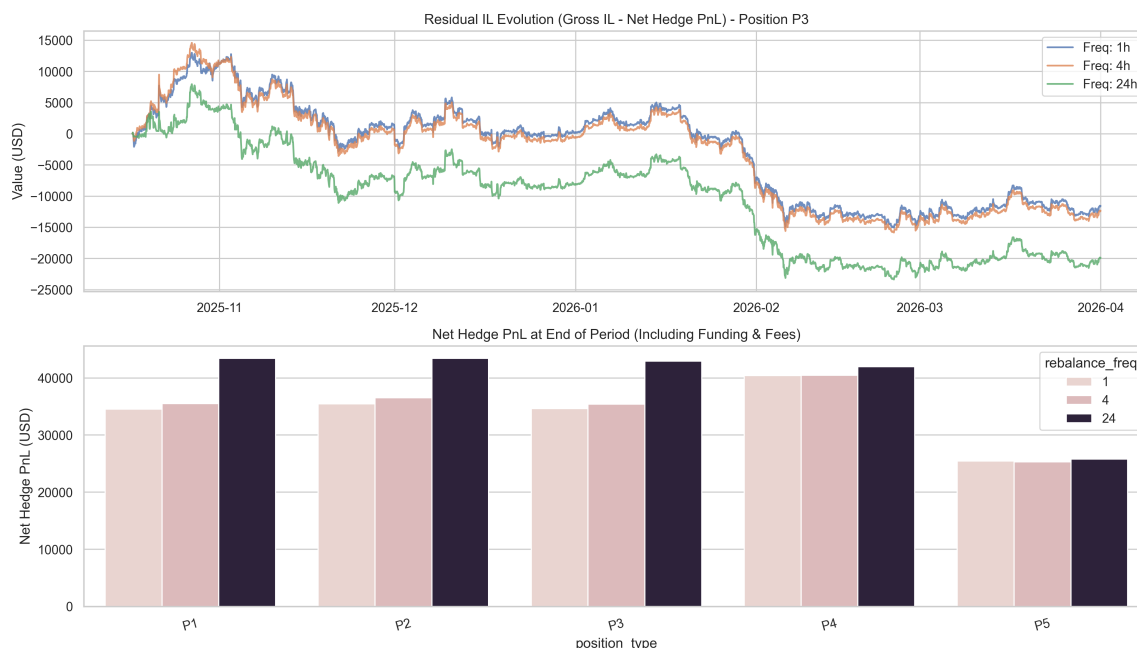


Figure 16: Net Hedging P&L vs. Gross IL across positions and rebalancing frequencies

Top Chart (Residual IL)

- Varies between $-\$25k$ and $+\$15k$ on a $\$100k$ notional = $\pm 15-25\%$ $\rightarrow$ significant but expected given the volatility.
- The hedge does not perfectly cover the IL (since we can observe a growing negative residue, particularly visible on the 24h frequency after Feb 2026) because ETH dropped by $\sim 55\%$, pushing narrow positions out of range and leaving large intra-day moves unhedged.
- Hedge tracking precision: $1h \approx 4h \gg 24h \rightarrow$ logical; frequent rebalancing leads to a tighter hedge and heavily limits the residual drift.

Bottom Chart (Net Hedge PnL)

- ETH dropped from $\sim \$4,466$ to $\sim \$2,100 = -53\%$.
- Shorting $\sim 2 \times 12 = 24$ ETH during this decline $\approx 24 \times (4466 - 2100) \approx \$56k$ for P3 (it is a bit less in reality because 2×12 is the extreme case when we are out of range all the time, which is not the case).
- **Important observation** For narrower positions (P1, P2, P3), the 24h frequency yields the *highest* Net Hedge PnL (up to $\sim \$43k$). This proves that the transaction costs and funding fees of 1h rebalancing eat heavily into the final profit.

Conclusion

Finally, we can say that this result is financially consistent since we shorted ETH during a major downturn, and the hedge performed well overall. The negative residue represents the uncovered IL when narrow positions went out of range.

Furthermore, the charts highlight a clear trade-off: high-frequency (1h) rebalancing minimizes the Residual IL (better risk coverage) but underperforms in Net PnL due to fee drag, whereas low-frequency (24h) rebalancing maximizes net profit despite a much looser, drifting hedge.

Task 5.4 — Limitations and Extensions

1. Residual Gamma Risk

The LP position has a continuously varying, negative Gamma, whereas the linear perp hedge has, by definition, zero Gamma.

The gap between the drifting LP Delta and the fixed hedge between each rebalancing creates a **residual Gamma risk** that grows with price volatility and the time interval (1h, 4h, 24h). The results clearly illustrate this dynamic:

- **1h and 4h frequencies:** Limit the drift, with the residual IL heavily fluctuating between +\$15,000 and −\$15,000;
- **24h frequency:** Drifts severely into negative territory, dropping as low as −\$25,000 due to the compounding effect of leaving large intra-day price moves unhedged.

Although narrower positions (like P1, P2) are theoretically more exposed to this risk due to their higher liquidity concentration (L), they exit their range almost immediately in practice. Therefore, residual IL depends more on the time spent in-range than solely on the range width.

2. Funding Rate Risk

Because the dynamic hedge alternates between short and long positions depending on price deviation, it avoids the structural funding bias of a permanent short hedge.

However, the ETH perp market experienced predominantly positive funding rates between October 2025 and March 2026. This means the hedge benefited from receiving funding during its short phases, artificially boosting profits. In a prolonged bear market with negative funding rates, the short side of the hedge would pay rather than receive, which would inevitably compress the net PnL.

3. Transaction Costs vs. Optimal Frequency

The 0.045% taker fee applied at every rebalancing heavily penalizes high-frequency execution.

- **Concentrated positions (P1, P2, P3):** As we can see on the graph, the 24h frequency significantly outperforms. The compounding costs of hourly rebalancing (fees and funding) destroy profitability, far outweighing the benefits of a tighter hedge.
- **Full-range position (P5):** Rebalancing frequency has virtually no impact. The range is so wide that adjustment trades are rarely triggered, rendering high-frequency execution unnecessary.

Conclusion: This strategy requires a **substantial minimum viable scale**. If the fee drag is already significant on a \$100k position at a 1h frequency, it would completely wipe out any hedging benefits on a smaller portfolio (e.g., \$10k).

4. The Theoretical Alternative: Options

Perps successfully hedge first-order exposure (Delta) but leave second-order exposure (Gamma) unhedged. Since the LP is structurally short Gamma, the ideal theoretical instrument is a **long Gamma position** - such as buying options symmetrically (e.g., a Straddle or Strangle - we've seen a straddle before). This would hedge Impermanent Loss without requiring continuous rebalancing.

But after asking if it is possible to Gemini, it seems complicated in practice since the crypto options ecosystem presents three major obstacles to this approach:

- **Fixed Expiries:** Traditional options require frequent, costly roll-overs that do not align with the continuous, path-dependent payoff of an LP position.
- **Theta (Time Decay):** The time cost of holding long options must be offset by LP fee income, which is never guaranteed—especially if the price drops out of range.
- **Bid-Ask Spreads:** Out-of-the-money (OTM) crypto options suffer from low liquidity and wide spreads, making precise replication economically unviable for anything below an institutional scale.

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